

S3 Pricing Model ←
(... , standard) storage class. 1

- 2. Number & size of objects
- 3. Data Transfer out S3 Region
- 4. Requests and data retrieval

RDS Pricing Model ←

- 1. Running clock hours
- 2. (memory, size, engine) Type of DB
- 3. storage
- 4. purchase type (Instance) (on-demand, RI...)
- 5. Database Gunit (number of instances)
- 6. API Requests & Calls (ما فيهم الـ calls (From insights dashboard)
- 7. Deployment type (single or multi AZs)

outbound data transfer

TCO خلاصنا الـ pricing models نندخل على الـ

(Total Cost of ownership) TCO ←

Financial estimate helps you to understand direct + indirect costs on AWS

لحساب الـ TCO (الحسبة المالية التقديرية) لما
ننقل migration على الكلاود استخدم
"Application Discovery Service"

3 ways to reduce TCO ←

1. Minimize Capex

2. Utilize Reserved Instance

3. Right size your resources

← انت حاليه خلاص عملت migration وعالوز estimates
AWS fees & charges based on your use case

وكان تعرف الـ instance type الـ fit الـ needs
عندك استخدم

"pricing calculator service" ←

الزن تون

[Pricing & Billing] Domain 4 ←

pricing ← حاجات بدفع
عليهم

1. (hourly) compute

2. storage

3. (outbound) data transfer

3 Free offer types ←

1. (initial sign up) 12 month free

2. always free

3. pay-as-you-go ← Trials
ما يجلس 30

EC2 Pricing Models ←

1. pay by hour (on-demand) (without pre paying)

2. saving plans (بتدفع مقدم على

compute usage

في الساعة لمدة 1 or 3 years

3. Reserved Instance (بتدفع مقدم لمدة

1 or 3 years ووافض النظر عن الـ

compute usage

4. Spot Instance (هتشتغل باقل سعر

فيه لوفقه متاح spare capacity

5. Dedicated Host (لو هتحتاج physical server
تولد الـ server

Lambda Pricing Models ←

1. Number of Requests (ما فيهم الـ test
or invoke

2. Execution time

3. Always free (1 million request
per month)

هنا الـ calculate بر دوبي
TCO على الـ time بتستخرج منه

Cost and Usage Report

يعني لو حاب في الامتحان
view granular data about AWS bill
يعني عرض ال data التفصيلية تفهم انه بيتكلم
عن ال Cost and Usage Report

تقدر تdownload ال report من عند
S3 Console

allow to visualize
and forecast your
costs and usage

يعرفك ال "visualize" ال data في
ال 12 month ال فاتوا و بيتنبأ (forecast)
ال Cost و ال usage برود ال 12 month
ال عاين

Analyze your EC2 Instances
usage over the past 7, 30

او لما في المدة بعد 12
شهر عادي تفهم ال Cost Explorer

useful for tracking
the spend (cost)

Via the Cost allocation report.

يعني اناب ال label ال resources المستخدمة
في نفس المشروع مثلا عشان بيتق
سهل اتابع ال Cost بتاعه

بعدما اتكلمنا عن ال Billings (الفواتير)
وال طرق المختلفة لعرض ال Cost على ال usage
فنتكلم حالا عن توحيد ال Billings ل
multiple AWS accounts
شركة واحدة او مشروع

ل ودا اسمه ال Governance

في طريقة ثانية انك تعرف سعر كل
service
حاليا من خلال query وهي استخدام
AWS Price List API

query using JSON or HTML
Receive price alerts when
prices change.

كل الفات دايك ال السعر pricing وعل
حسبة تكاليف تقديرية وهكذا ...

انا بقى حالا علوزه اعرف انادفعت
كام او علما كام في الفاتورة (Billing)

track your ongoing spend
Billing

alert you when
exceed your budgeted
amount (تتجاوز ما
لقد ال threshold)

تقدر تقيس budget على ال Cost
usage
ال حجز بقر بخص
reservation او تخطيط
savings plans

من ال Real World Case
انك تقبل budget بتا كدائك
معدتي ال free tier limits

provides the most detailed
set of AWS
Cost and Usage
data

يعطيك تفاصيل عن الاستخدام
ال انت دفعت عليه التكلفة دي

في الاصل بيوفر ال usage على كل service
Category list
وتقدر ت aggregate ال usage على نفسه وبيوفر ال
daily or hourly or monthly

Benefits ثانيه من استخدام المنظمات

Automate account creation

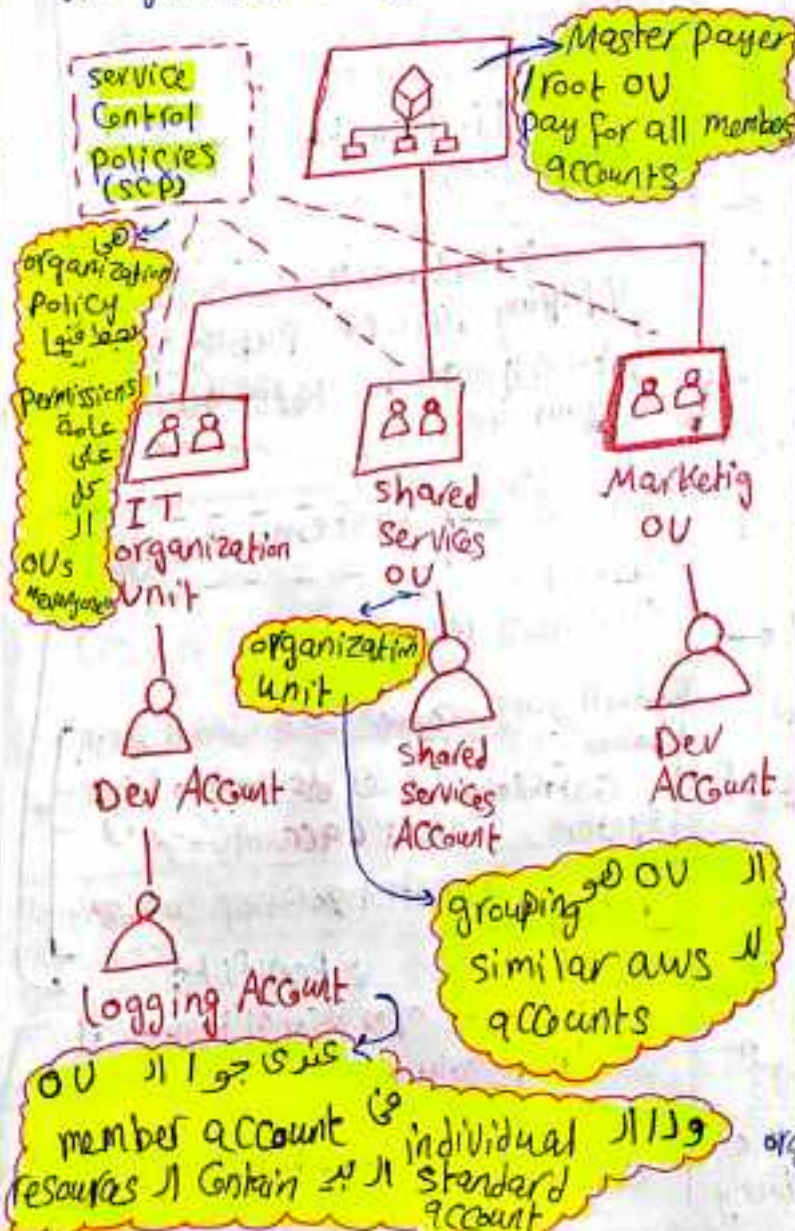
لقد امكن خلال api متاح بيقها
ورا حلوجدا لو هيقبل account
مثلا Sandbox وعاوز خط عليه
Restrictions
كثيره فدي افضل واسرع طريقة انك
تعمله
Creation كذا، تقدر تمانت
invite existing account Programmatically



Restrict account privileges using Service Control Policies (SCP)

لنشر جوار لوقتي
allocate resources and apply policies

Structure بنوع المنظمات



Governance

allow you to manage multiple accounts under one umbrella.

Global Service
بتسمحلك
اكثر من اكونت في وقت واحد

main account
الباقي

The master account

طب اسي الفائدة (Cost benefits)
من المنظمات

single payment for all accounts
[Consolidated Billing]

2. receive volume discounts since usage is combined across accounts.

معني انك عملت حاجة دلوقتي
اسمها aggregated usage
بتستولك كم كبير من ال resources
تحت مسمي اكونت واحد
هيد بلك خصم كبير (جفتهم)

3. Reduce costs by sharing resources across accounts

فلوانت عندك RI Instance
محجوزة لمدة سنة او ٣ سنين عند
اكونت فيهم لكنه مش بيستخدمها
حالا، فبقدر اكونت ثاني في المنظمة
بيستفيد منها وبيستخدمها هو

3. patch & run commands on multiple EC2 instances or RDS instances

← والخدمة دي جاية عليها سويفت الامثلة
software و deploy OS patches
large group of instances
فالإحالة باستخدام الـ
• System Manager

4. Trusted Advisor
← مساعدة على high level account assessment

← provides real-time guidance to help provision your resources following AWS best practices.

[Service بتعمل checks (assessment) على
الأكاؤنت بتاعك في حاجات ليها علاقة
بـ performance, security, ...]

← Benefits

1. checks your account and makes recommendations.

2. Help you to see services limits

3. Help you to understand best practices

← recommendations ال بيديها لك بتكون

منقسمة إلى 5 categories

1. Cost optimization
← delete unused resources

Free checks ال فينولوس

Business & Enterprise support plan
← هي فقط متاحة لـ

2. Performance
← Business و Enterprise

← check مهمة ببيتأكد منها وفي
checks for CloudFront content delivery optimization

2. Control Tower
← help to set up new accounts using multi account strategy

→ works directly on AWS organizations

→ Enforce the best use of services across accounts

→ provides dashboard to manage accounts.

← هي خدمة قائمة على AWS organizations
بعتي انها بتعمل organizations برودو
لـ automatic واسع والتالي
هتأكد ان accounts conform to
company policies
SCP implement لـ automatically

← استخدامها في الـ real world زي
applying disallow public write
to all S3 buckets across
your accounts

3. System Manager
← Give you control over AWS resources

← الاثنين ارفاقوا الـ AWS organizations و الـ Control Tower
كانوا ماشيين على نظام الـ controlling system
لـ accounts و الـ manager resources
Control & Group

← Benefits

1. Automate operational tasks on resources

2. Group resources to take action

Basic & Free Developer ← Service Limits • 5

Check for service usage greater than 80% over service limit.

Cost optimization → Check read and write capacity service limits for DynamoDB

Trusted Advisor helps reduce your overall costs by monitoring service limits.

Help you to manage software License. ← License Manager • 5

Controlling tracking license application running on EC2 instance
Compliance + agreement of license

: Benefits

Manage on-premises & AWS Licenses

Track licenses for Oracle, Microsoft, SAP -- (Vendor license)

Help ← AWS Certificate Manager (ACM) • 6
provisioning and manage SSL/TLS Certificate

Generate Certificate for SSL/TLS access
http to https

Recommendation Categories ← Security • 8

Free Basic & Developer support plan

checks for unrestricted access for specific port on EC2 instance

checks for S3 bucket permissions to determine if public access

checks for multi-factor authentication (MFA) on root account

checks for RDS public snapshots

Free Business & Enterprise support plans

checks IAM Password Policy

checks for exposed access keys

Free Business & Enterprise support plans ← Fault tolerance • 4

Management Services ✖

Amazon Managed Services (AMS)¹

↳ is actually a team of people that provides you infrastructure and application support on AWS

Team of AWS Service experts (AMS team) operate & manage your infrastructure security, reliability and availability

: Benefits

Augments your internal staff
← يزيد عدد موظفيك الداخليين
management

Provides ongoing of your infrastructure
← إدارة مستمرة
للبنية التحتية
باعتك

Reduce operational risks and overhead

← هذا بيوفرلك (ببساطة):
Patch management, Monitoring,
Cost optimization

وسغالين 24/365 hours

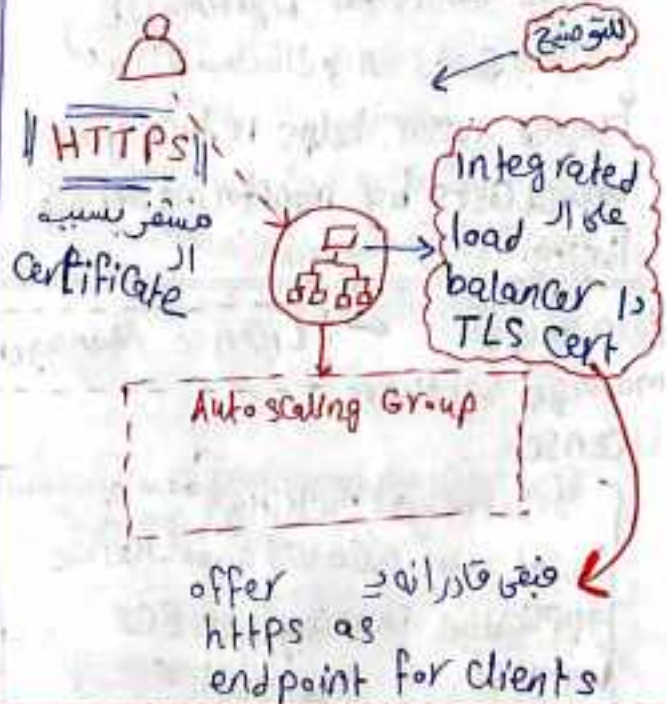
← باب مثال ج في exam قبل كذا

Managed Services can increase your operational efficiency by helping you develop application-specific health monitoring using CloudWatch.

Certificate Manager (ACM) ← : Benefits

Provides public & private certificates for free
← ويبعد Automatic

Integrated with Elastic Load Balancer, API Gateway, CloudFront distributions



← بعد ال Governance والسياسة في ال accounts وال resources وكل حاجة، دلو قتي

Management Services
التي بيكون إدارتها و AWS ومن ال best practices
utilization، لا غاب support you

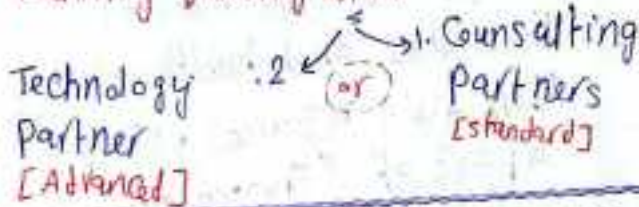
Planned Events -
Product Launches -
Migrations -

Benefits

1. offers technology partners that provides software solutions, (مقدماء (شركاء) تقنيين)
2. offers / provides Consulting partners that offers professional services, (استشارات تخصص ال solution وكذا)
3. Find approved vendors with deep AWS expertise. (مختلطين بعمق شركات معتمدة عندها خبرات عميقة وكبيرة فيما يخص AWS)

مثال جوهني امقان

Team lacks the technical expertise to build and deploy Cloud application, the APN could help you get up and running quickly with



AWS Marketplace

مقدماء افانوا كانوا people, اسفاد ابقى
 → is online platform provided by AWS where customers can deploy software products and services that can run on AWS infrastructure.

Marketplace is a digital catalog of prebuilt solutions you can purchase or license, you may also sell your own solutions to others

Professional services

team of people (فريق من الناس)
 يسالفق انه ال AMS كان
 operate و manage ال infrastructure (يعمل بخدماتك)
 ينزل بخدماتك و يحمي ال security
 وما يقى ال professional services هو خاص بال enterprise customers
 ليساعدهم على الانتقال ال Cloud (helps enterprise customers move to cloud-based operating model)
 طب و هيفلدا ازاى؟

proposes solution

بمقدماء (يقترح على solution)
 مفيين اوسكل مفيين ال infrastructure migration على ال Cloud (quickly move to Cloud from on-premises)

Architects solutions

بمقدماء و فلا

Implement solutions

بمقدماء ببقى ال architecture ال رسمه

AWS Partner Network (APN)

هو global community approved partners (بمقدماء شركاء معتمدين)
 بمقدماء Software solutions و يقدموا
 كمان استشارات (consulting services)
 فيما يتعلق ب AWS

* Support Plans

يعني بوفر للـ AWS customer نوع معين من الـ technical assistance
مشاركة بناء على الـ support plan
الذي مشترك فيها [نظرة على الـ support plan]

← عذري 4 Support Plans

① Basic support plan

② Is included for free for

③ all AWS accounts

④ : Features

1. Account and billing support cases

2. service limit increase

3. access to technical support

الوحيدة المتاحة هي customer service via email 24/7

← Response Time

← response time

• technical support

2. Developer support plan

Developer support starts at 29 \$/month

and is recommended for testing and development.

← : Features

1. Account and billing cases

2. service limit increases

← : Marketplace Benefits

1. Buy third-party software

2. Sell solutions to other customers

3. search the Catalog of software listings and install with the click of a button

4. Try out the app before making a long-term commitment (Free trial)

5. Personal Health Dashboard

(PHD)

is a feature provided by AWS to help customers monitor the status and health of their AWS resources, it's a part of "Trusted Advisor"

Benefits

1. Personal Health Dashboard

alerts you to events that might impact your AWS environment

• notify you of events that might impact your AWS environment

• provides troubleshooting guidance

• tailored advice

• Feedback tailored to your specific environment

Business plan Technical support

unlimited Contacts .1

unlimited Cases .2

Full set of Trusted Advisor .3
Checks

cloud support Engineers .4
التواصل سيكون مع

[24/7 access via email, phone or chat]

Response Time

< 24 hours General guidance .1

< 12 hours system impaired .2

< 4 hours Production system impaired .3

< 1 hours Production system down .4

Enterprise Support Plan .4

Enterprise support starts at 15,000 \$ month and is recommended for business or mission-critical production workloads.

Features

Account and billing cases .1

Service limit increases .2

Technical support .3

Developer support plan Features (تابع ...)

Technical support .3

1 Primary Contact .1

ستكون واحد هو يكون
وسيلة التواصل مع الدعم
Team

Unlimited Cases .2

الدعم التواصل مع الدعم
التي عدد من المرات ...

cloud support .3
التواصل سيكون مع

Associate in [Business hours
via email only]

Response Time

< 24 hours General guidance .1

< 12 hours system impaired .2

Business Support Plan .3

Business support plan starts at 100 \$ month and is recommended for production workloads.

Features

Account and billing cases .1

Service limit increases .2

Technical support .3

< 4 hours production system impaired .3

< 1 hours production system down .4

< 15 minutes Business-critical system down .5

Enterprise plan Technical support

Unlimited contacts .1

Unlimited cases .2

Full set of Trusted Advisor checks .3

Technical Account Manager (TAM) .4

TAM is your designated contact with AWS, he monitor your environment and provide [best practices guidance and assistance with architecting.]

Concierge support team .5

[billing and account questions]

Infrastructure Event Management .6

[Operational support
• migrations launches]

Response Time

< 24 hours General guidance .1

< 12 hours System impaired .2